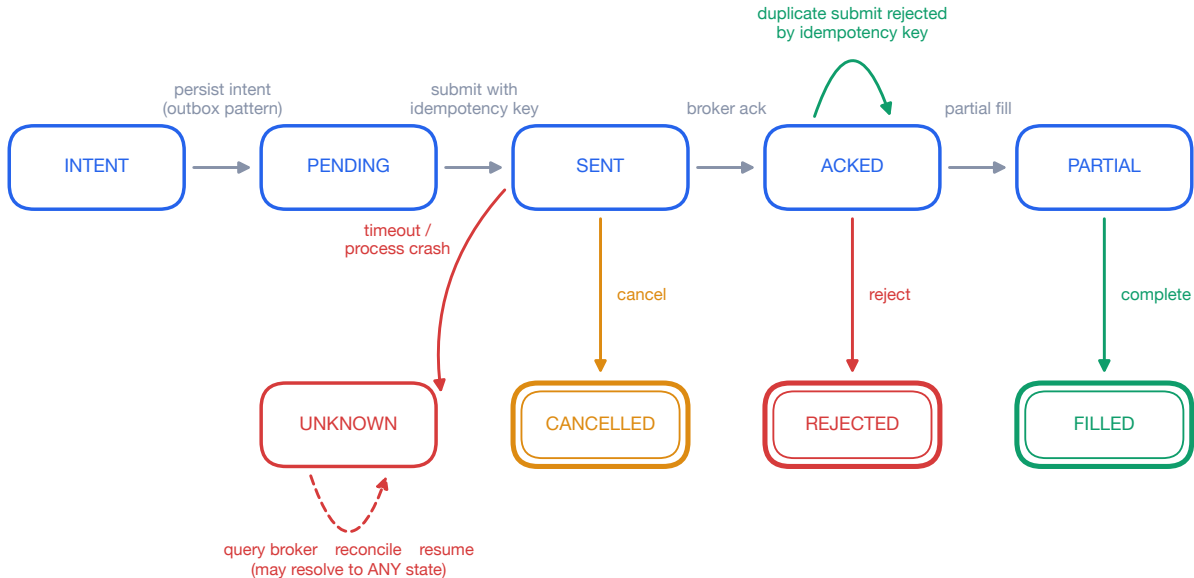


Figure 10 — Order lifecycle: every transition persisted, every submission idempotent



Double-ruled states are terminal. UNKNOWN is the state most systems omit and the one that causes duplicate orders: never assume an unacknowledged order was not filled. This machine is the natural target for the TLA+ model-checking and Hypothesis stateful testing recommended in §6.1.